

Derivatives

Gradient

$$\text{grad } f = \nabla f = [f_x, f_y, f_z]$$

Directional Derivative

$$D_{\mathbf{a}} f = \frac{df}{ds} = \frac{\mathbf{a}}{|\mathbf{a}|} \cdot \nabla f$$

Divergence

$$\text{div } \mathbf{v} = \nabla \cdot \mathbf{v} = \frac{\partial v_1}{\partial x} + \frac{\partial v_2}{\partial y} + \frac{\partial v_3}{\partial z}$$

$$\nabla \cdot \mathbf{v} = 0 \rightarrow \text{incompressible}$$

Curl

$$\text{curl } \mathbf{v} = \nabla \times \mathbf{v} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ v_1 & v_2 & v_3 \end{vmatrix}$$

Laplacian

$$\Delta f = \nabla \cdot (\nabla f) = \nabla^2 f = f_{xx} + f_{yy} + f_{zz}$$

Common Equalities

$$\nabla f g = f \nabla g + g \nabla f$$

$$\nabla \frac{f}{g} = \frac{g \nabla f - f \nabla g}{g^2}$$

$$\nabla \cdot (f \mathbf{v}) = f \nabla \cdot \mathbf{v} + \mathbf{v} \cdot \nabla f$$

$$\nabla \cdot (f \nabla g) = f \Delta g + \nabla f \cdot \nabla g$$

$$\Delta(fg) = f \Delta g + 2 \nabla f \cdot \nabla g + g \Delta f$$

$$\nabla \times (f \mathbf{v}) = \nabla f \times \mathbf{v} + f(\nabla \times \mathbf{v})$$

$$\nabla \cdot (\mathbf{u} \times \mathbf{v}) = \mathbf{v} \cdot (\nabla \times \mathbf{u}) - \mathbf{u} \cdot (\nabla \times \mathbf{v})$$

$$\nabla \times (\nabla f) = \mathbf{0}$$

$$\nabla \cdot (\nabla \times \mathbf{v}) = 0$$

Normal Vector

$$\text{Case } f(x, y, z) = \text{const: } \mathbf{N} = \nabla f$$

$$\text{Case } z = f(x, y): \mathbf{N} = [f_x, f_y, -1]$$

Tangent Plane

At point (x_0, y_0, z_0) , the tangent plane is

$$N_x(x - x_0) + N_y(y - y_0) + N_z(z - z_0) = 0$$

Line Integrals

$$\text{Arc Length } ds = |\mathbf{r}'(t)| dt$$

$$\int_C \mathbf{F}(\mathbf{r}) \cdot d\mathbf{r} = \int_a^b \mathbf{F}(\mathbf{r}(t)) \cdot \mathbf{r}'(t) dt$$

$$\int_C (F_1 dx + F_2 dy + F_3 dz) = \int_a^b (F_1 x' + F_2 y' + F_3 z') dt$$

The line integral is **path independent** iff,

1. Field is **conservative**: $\mathbf{F} = \nabla f$
2. Value around all closed paths is **zero**
3. Field is **exact**: $\nabla \times \mathbf{F} = \mathbf{0}$

Note: If $\mathbf{F}$ is defined on a *simply connected domain* (i.e. no holes), then $\mathbf{F}$ is conservative $\iff \mathbf{F}$ is exact

If path independent,

$$\int_A^B \mathbf{F} \cdot d\mathbf{r} = \int_A^B \nabla f \cdot d\mathbf{r} = f(B) - f(A)$$

Finding f such that $\mathbf{F} = \nabla f$

1. Verify the field is exact.
2. Integrate one component of $\mathbf{F}$ wrt the corresponding variable.
The result is a fn plus an unknown fn of the other variables.
Ex.

$$f = \int F_1 dx + g(y, z)$$

3. Take the partial of the next variable and equate to corresponding component.
4. Rinse & repeat. If functions turn out to be a constant, choose zero.

Green's Theorem

$$\iint_R \nabla \times \mathbf{F} \cdot \mathbf{k} dA = \oint_C \mathbf{F} \cdot d\mathbf{r}$$
$$\iint_R \left(\frac{\partial F_2}{\partial x} - \frac{\partial F_1}{\partial y} \right) dx dy = \oint_C (F_1 dx + F_2 dy)$$

Area Formulas

$$\text{Area} = \oint_C x dy$$

$$\text{Area} = - \oint_C y dx$$

$$\text{Area} = \frac{1}{2} \oint_C (x dy - y dx)$$

$$\text{Area} = \frac{1}{2} \oint_C r^2 d\theta$$

Transforming the Laplacian

$$\iint_R \Delta w dA = \oint_C \nabla w \cdot \mathbf{n} ds = \oint_C \frac{\partial w}{\partial n} ds$$

Surfaces

$$\mathbf{r} = \mathbf{r}(\mathbf{u}, \mathbf{v})$$

$$\mathbf{N} = \mathbf{r}_u \times \mathbf{r}_v$$

$$\text{Cylinder } x^2 + y^2 = a^2, \quad |z| \leq 1$$

$$\mathbf{r}(z, \theta) = \begin{bmatrix} a \cos \theta \\ a \sin \theta \\ z \end{bmatrix}, \quad \theta \in [0, 2\pi]$$

$$\text{Sphere } x^2 + y^2 + z^2 = a^2$$

$$\mathbf{r}(\phi, \theta) = \begin{bmatrix} a \sin \phi \cos \theta \\ a \sin \phi \sin \theta \\ a \cos \phi \end{bmatrix}, \quad \begin{array}{l} \phi \in [0, \pi] \\ \theta \in [0, 2\pi] \end{array}$$

Cone $z = \sqrt{x^2 + y^2}$

$$\mathbf{r}(r,\theta) = \begin{bmatrix} r \cos \theta \\ r \sin \theta \\ r \end{bmatrix}, \quad \begin{array}{l} r \in [0,H] \\ \theta \in [0,2\pi] \end{array}$$

Torus

With major radius R , and minor radius r

$$\mathbf{r}(\phi,\theta) = \begin{bmatrix} (R+r\cos\phi)\cos\theta \\ (R+r\cos\phi)\sin\theta \\ r\sin\phi \end{bmatrix}, \quad \begin{array}{l} \phi \in [0,2\pi] \\ \theta \in [0,2\pi] \end{array}$$

Surface Integrals

Area element $dA = ||\mathbf{r}_u \times \mathbf{r}_v|| \; du \; dv = ||\mathbf{N}|| \; du \; dv$

Flux

The amount of fluid passing through a surface S per unit time. S can also be parameterized as $\mathbf{r}(u,v)$ for $(u,v) \in \Omega$.

$$\text{Flux} = \iint_S \mathbf{F} \cdot \mathbf{n} \; dA = \iint_\Omega \mathbf{F}(\mathbf{r}) \cdot ||\mathbf{N}|| \; du \; dv$$

Surface Area

$$\text{Surface Area} = \iint_\Omega ||\mathbf{N}|| \; du \; dv$$

Scalar Surface Integral

$$\iint_S G \; dA = \iint_\Omega G(\mathbf{r})||\mathbf{N}|| \; du \; dv$$

Divergence Theorem

Let V be a solid region with boundary S oriented outward. If $\mathbf{F}$ is continuously differentiable,

$$\iiint_V \nabla \cdot \mathbf{F} \; dV = \iint_S \mathbf{F} \cdot \mathbf{n} \; dA$$

Stokes' Theorem

Let S be an oriented surface with right-hand boundary C . If $\mathbf{F}$ is continuously differentiable,

$$\iint_S \nabla \times \mathbf{F} \cdot \mathbf{n} \; dA = \oint_C \mathbf{F} \cdot d\mathbf{r}$$

Triple Integrals

Rectangular: $dV = dx \; dy \; dz$

Cylindrical: $dV = r \; dr \; d\theta \; dz$

Spherical: $dV = \rho^2 \sin \phi \; d\rho \; d\theta \; d\phi$

Green's Identities

Derived by applying the Divergence Theorem to the vector field $\mathbf{F} = f \nabla g$.

Green's First Identity

$$\iiint_T (f \Delta g + \nabla f \cdot \nabla g) \; dV = \iint_S f \frac{\partial g}{\partial n} \; dA$$

Green's Second Identity

$$\iiint_T (f \Delta g - g \Delta f) \; dV = \iint_S \left(f \frac{\partial g}{\partial n} - g \frac{\partial f}{\partial n} \right) \; dA$$

Trigonometry

Identities

$$\sin^2 \theta = \frac{1}{2} - \frac{1}{2} \cos 2\theta$$

$$\cos^2 \theta = \frac{1}{2} + \frac{1}{2} \cos 2\theta$$

$$\sin \theta \cos \theta = \frac{1}{2} \sin 2\theta$$

$$\sin(\alpha + \beta) = \sin \alpha \cos \beta + \cos \alpha \sin \beta$$

$$\sin(\alpha - \beta) = \sin \alpha \cos \beta - \cos \alpha \sin \beta$$

$$\cos(\alpha + \beta) = \cos \alpha \cos \beta - \sin \alpha \sin \beta$$

$$\cos(\alpha - \beta) = \cos \alpha \cos \beta + \sin \alpha \sin \beta$$

Values

	0	$\pi/6$	$\pi/4$	$\pi/3$	$\pi/2$
$\sin \theta$	0	1/2	$\sqrt{2}/2$	$\sqrt{3}/2$	1
$\cos \theta$	1	$\sqrt{3}/2$	$\sqrt{2}/2$	1/2	0
$\tan \theta$	0	$1/\sqrt{3}$	1	$\sqrt{3}$	—